Trolls, Haters, or Teammates: PvP Chat Sentiment & Toxicity Classification

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Course: UE24CS352A — Machine Learning Mini-Project

1. Abstract & Introduction

In competitive player-versus-player (PvP) online gaming environments, real-time text communication is essential for team coordination. However, toxicity, hostile banter, and derogatory language frequently degrade user experience and player retention. This project presents a machine learning pipeline designed to process unstructured PvP game chat messages and classify them into distinct sentiment categories: **Positive**, **Neutral**, and **Derogatory / Toxic**.

2. Dataset & Preprocessing

The dataset consists of multi-class gaming chat entries representing common in-game expressions, tactical commands, and toxic statements.

- **Text Normalization:** Lowercasing, noise removal, and tokenization.
- **Feature Extraction:** Term Frequency-Inverse Document Frequency (TF-IDF) vectorization to capture word importance and n-gram representations unique to gaming jargon (e.g., *"gg EZ"*, *"trash team"*).

3. Methodology & Model Architecture

We evaluated multiple classification architectures to identify the optimal balance between prediction accuracy and real-time inference latency required for live in-game moderation:

1. **Baseline Classifiers:** We implemented K-Nearest Neighbors (K-NN) and Multinomial Naive Bayes operating on Term Frequency-Inverse Document Frequency (TF-IDF) feature vectors. TF-IDF was selected to effectively capture word relevance and n-gram representations unique to gaming jargon (e.g., *"gg EZ"*, *"trash team"*).
2. **Sequence Modeling:** We evaluated a Bidirectional LSTM (Bi-LSTM) sequence network initialized with 50-dimensional GloVe embeddings to capture complex context and sentiment intent across conversational chat phrasing.
3. **Pipeline Integration:** Model weights (model.pkl) and text processing transformers (vectorizer.pkl) were serialized using pickle and integrated into a real-time Streamlit web app (demo.py) for live prediction.

4. Experimental Results & Performance Analysis

- **Classification Accuracy:** The Naive Bayes and Bi-LSTM models demonstrated strong class separation across multi-class gaming messages (Positive, Neutral, Derogatory / Toxic).
- **Confidence Scores:** The model achieved over 90% confidence on core toxic inputs (e.g., *"gg EZ trash team uninstall"*).
- **Confusion Matrix Analysis:** The model effectively minimized false positives, cleanly separating neutral tactical orders (e.g., *"defend top tower"*, ~51.4% neutral confidence) and positive team cheerleading (e.g., *"nice play dude gg"*, ~51.6% positive confidence) from actual derogatory remarks.

5. Conclusion & Deployment

The deployed Streamlit demo validates that lightweight machine learning models can deliver high-confidence toxicity detection suitable for integration into automated moderation pipelines.